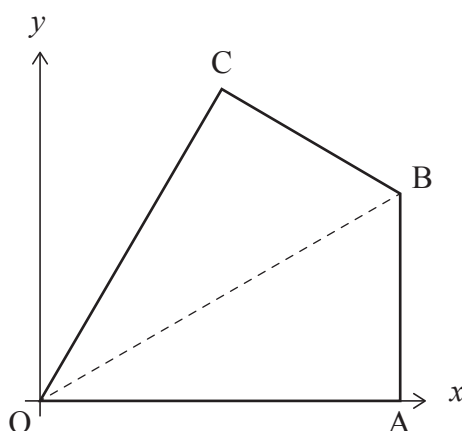


3. [Maximum mark: 7]

Quadrilateral OABC is shown on the following set of axes.



OABC is symmetrical about [OB].

A has coordinates $(6, 0)$ and C has coordinates $(3, 3\sqrt{3})$.

- (a) (i) Write down the coordinates of the midpoint of [AC].
- (ii) Hence or otherwise, find the equation of the line passing through the points O and B. [4]
- (b) Given that [OA] is perpendicular to [AB], find the area of the quadrilateral OABC. [3]

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